

Markov Chain

Joint distrⁿ

$$P(\underbrace{X_1=i_1}_{A_1}, \dots, \underbrace{X_n=i_n}_{A_n} \mid X_0=i_0)$$
$$= p_{i_0 i_1} p_{i_1 i_2} \dots p_{i_{n-1} i_n}.$$

$$P(A_n \cap A_{n-1} \cap \dots \cap A_1)$$
$$= P(A_n \mid A_{n-1} \cap \dots \cap A_1)$$
$$\times \dots \times P(A_2 \mid A_1) P(A_1)$$



$$P(X_n = j \mid X_0 = i)$$

Notation:

$$P_{ij}^{(n)} = P(X_n = j \mid X_0 = i)$$

$$\begin{aligned}
 P_{ij}^{(2)} &= P(X_2 = j \mid X_0 = i) = \sum_{k \in S} P(X_2 = j, X_1 = k \mid X_0 = i) \\
 &= \sum_{k \in S} P_{ik} P_{kj}
 \end{aligned}$$

$$P_{ij}^{(2)} = \sum_{k \in S} P_{ik} P_{kj} = (P^2)_{ij} \quad P = ((P_{ij}))$$

$$A = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_i \\ \vdots \\ a_n \end{pmatrix} \quad B = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_i \\ \vdots \\ b_n \end{pmatrix}$$

$$P^2 = \left(\left(p_{ij}^{(2)} \right) \right)$$

$$P^2 \underline{1}' = P \cdot P \underline{1}'$$

$$= P \cdot \underline{1}$$

$$= \underline{1}$$

Entries are non-neg
 $\sum_{j \in S} p_{ij}^{(2)} = 1$

Ex: Show all not all st. matrices
 can arise as P^2 .

Claim $P^n = \left(\left(P_{ij}^{(n)} \right) \right) \quad \forall n \geq 1$

Assume this for n .

$$P_{ij}^{(n+1)} = P(X_{n+1}=j | X_0=i)$$

$$\begin{aligned} P(A \cap B | C) \\ = P(A | B \cap C) \\ \times P(B | C) \end{aligned}$$

$$= \sum_{k \in \mathcal{S}} P(X_{n+1}=j, X_n=k | X_0=i)$$

$$= \sum_{k \in \mathcal{S}} \frac{P(X_{n+1}=j | X_n=k, X_0=i)}{P(X_n=k | X_0=i)}$$

$$= \sum_{k \in S} P(X_{n+1}=j | X_n=k) P(X_n=k | X_0=i)$$

$$= \sum_{k \in S} \underbrace{P_{ik}^{(n)}}_{(P^n)_{ik}} P_{kj}$$

$$= (P^{n+1})_{ij}$$

P^n is also stochastic matrix.

$$P^n = \left(\left(p_{ij}^{(n)} \right) \right) \quad \forall n \geq 1.$$

For $m, n \geq 1$,

$$P^{m+n} = P^m P^n = \left(\left(p_{ij}^{(m)} \right) \right) \left(\left(p_{ij}^{(n)} \right) \right)$$

i.e.,

$$p_{ij}^{(m+n)} = \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)}$$

Set $p_{ij}^{(0)} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{o.w} \end{cases} = \delta_{ij}$

$P^0 = I$. Then, $P^{m+n} = P^m P^n$

Chapman Kolmogorov Eqⁿ. $\forall m, n \geq 0$

For $i, j \in S$, $m, n \geq 0$,

$$p_{ij}^{(m+n)} = \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)}$$

Example :- $\{X_n: n \geq 0\}$ iid.

$$P = P^n \quad P(X_n = j \mid X_0 = i) = P(X_n = j) \\ = \alpha_j$$

$$P^n = \begin{pmatrix} & -1 & 0 & 1 \\ & : & & \\ \cdot & \cdot & \alpha_{-1} \alpha_0 \alpha_1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & \alpha_{-1} \alpha_0 \alpha_1 & \cdot & \cdot & \cdot \\ & \alpha_{-1} \alpha_0 \alpha_1 & \cdot & \cdot & \cdot & \cdot \end{pmatrix} \\ = P(X_1 = j \mid X_0 = i)$$

Ex:- Two state MC. $\mathcal{S} = \{0, 1\}$.

$$P = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix}$$

$$P^n = \begin{pmatrix} p_{00}^{(n)} & p_{01}^{(n)} \\ p_{10}^{(n)} & p_{11}^{(n)} \end{pmatrix}$$

$$p_{00}^{(n+1)} = \sum_{k \in \mathcal{S}} p_{0k}^{(n)} p_{k0}$$

$$= p_{00}^{(n)} p_{00} + p_{01}^{(n)} p_{10} = p_{00}^{(n)} p_{00} + (1 - p_{00}^{(n)}) p_{10}$$

$$p_{00}^{(n+1)} = p_{10} + p_{00}^{(n)} (p_{00} - p_{10})$$

$$\begin{aligned}
 &= p_{10} + (p_{00} - p_{10}) \left[p_{10} + p_{00}^{(n-1)} (p_{00} - p_{10}) \right] \\
 &= p_{10} + p_{10} (p_{00} - p_{10}) + (p_{00} - p_{10})^2 p_{00}^{(n-1)} \\
 &\vdots \\
 &= p_{10} + p_{10} (p_{00} - p_{10}) + \dots + p_{10} (p_{00} - p_{10})^n \\
 &\quad + (p_{00} - p_{10})^{n+1} p_{00}^{(0)}
 \end{aligned}$$

Assume
 $0 < p_{ij} < 1$

$$p_{00}^{(n+1)} = p_{10} + p_{10}(p_{00} - p_{10}) + \dots + p_{10}(p_{00} - p_{10})^n$$

$$+ (p_{00} - p_{10})^{n+1}$$

$$= p_{10} \left[\frac{1 - (p_{00} - p_{10})^{n+1}}{1 - (p_{00} - p_{10})} \right] + (p_{00} - p_{10})^{n+1}$$

$$p_{00}^{(n)} = \frac{p_{10}}{p_{10} + p_{01}} - (p_{00} - p_{10})^n \left[1 - \frac{p_{10}}{p_{10} + p_{01}} \right]$$

$$= \bar{J}l_0 + \lambda_2 C_{00}$$

Conclusion:

$$p_{00}^{(n)}$$

$$\rightarrow \bar{p}_0$$

$$= \frac{p_{01}}{p_{01} + p_{10}}$$

$$\lambda_2 = p_{00} - p_{10}$$

$$|\lambda_2| < 1$$

$$p_{01}^{(n)} = 1 - p_{00}^{(n)}$$

$$\rightarrow 1 - \bar{p}_0 = \bar{p}_1 = \frac{p_{10}}{p_{01} + p_{10}}$$

$$P = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix},$$

Second eigenvalue =

$$p_{00} + \underbrace{p_{11} - 1}_{\lambda_2} = p_{00} - p_{10} = \lambda_2.$$

$$|p_{00}^{(n)} - \pi_0| \leq C |\lambda_2|^n$$

$$p_{01}^{(n)} = 1 - p_{00}^{(n)}$$

$$, \quad \pi_1 = \frac{p_{10}}{p_{10} + p_{01}}$$

$$= \pi_1 + \lambda_2^n C_{01}$$

$$\Rightarrow |p_{01}^{(n)} - \pi_1| \leq C' |\lambda_2|^n$$

$$p_{11}^{(n)} = \frac{p_{01}}{p_{01} + p_{10}} + (p_{11} - p_{01})^n \left(1 - \frac{p_{01}}{p_{01} + p_{10}} \right)$$

$$= \pi_1 + \lambda_2^n C_{11}$$

$$\rightarrow p_{11}^{(n)} \rightarrow \pi_1$$

$$\Rightarrow p_{10}^{(n)} = 1 - p_{11}^{(n)} \rightarrow \pi_0$$

$$\begin{pmatrix} p_{00}^{(n)} & p_{01}^{(n)} \\ p_{10}^{(n)} & p_{11}^{(n)} \end{pmatrix} \longrightarrow \begin{pmatrix} \pi_0 & \pi_1 \\ \pi_0 & \pi_1 \end{pmatrix}$$

$$\underline{\pi} = (\pi_0 \ \pi_1)$$

Check $\underline{\pi} P = \underline{\pi}$

$$\underline{\pi} \mathbf{1}' = 1.$$

$$\pi P = \pi \rightarrow$$

Balance eqⁿ.

π is called a
Stationary distrⁿ.

$$P(X_0=0) = \pi_0, \quad P(X_0=1) = \pi_1$$

$$\begin{aligned} P(X_1=0) &= P(X_1=0 | X_0=0) P(X_0=0) \\ &\quad + P(X_1=0 | X_0=1) P(X_0=1) \\ &= \pi_{00} \pi_0 + \pi_{10} \pi_1 \end{aligned}$$

$$P(X_1=1) = \pi_1$$